Dosage Calculation & IV Drip Rates

Formulas, worked examples and common traps for nursing medication math and NCLEX review

MindShelf

Drugs A–F and their numbers: exercise values, not dosing guidance.

1 Core methods

Three methods, one answer
Desired over have: $V = D/H \times Q$. D dose ordered, H dose on hand, Q volume (or tablets) holding H. **Ratio–proportion:** $H : Q = D : V$, then cross-multiply. **Dimensional analysis:** chain the order through conversion factors until only the wanted unit remains.

EXAMPLE Injection, three ways (Drug A)
Drug A ordered 28 mg; vial 32 mg per 1 mL.
D/H × Q: 28 ÷ 32 × 1 = 0.875 ⇒ **0.88 mL**
Ratio 32 : 1 = 28 : V and DA agree. Use a 1 mL syringe (0.01 mL marks).

TRAP
Divide only once units match: g against mg is off by a factor of 1,000.

2 Unit conversions

The ×1,000 ladder
g → mg → mcg, L → mL: × 1,000 per step down, ÷ per step up. lb → kg: × 0.45359237 (exact; courses ÷ 2.2). in → cm: × 2.54 (exact).

US measure	NIST (exact, rounded)	Label & course
1 tsp	≈ 4.93 mL	5 mL
1 tbsp	≈ 14.79 mL	15 mL
1 fl oz	≈ 29.57 mL	30 mL
1 cup	≈ 236.6 mL	240 mL

EXAMPLE Child's weight
44 lb × 0.45359237 ≈ 19.958 ⇒ **20 kg** (to 0.1 kg); course: 44 ÷ 2.2 = 20 kg.

CLINICAL NOTE
Oral liquids: dose in mL with an oral syringe, never a kitchen spoon.

TRAP
Pounds typed as kg: the dose is about 2.2 times too high.

3 Weight-based & safe dose

dose = mg/kg × kg; **per day** ÷ doses per day = **per dose**. Safe if low end × kg ≤ ordered daily total ≤ high end × kg.

EXAMPLE Safe-range check (Drug B)
17–33 mg/kg/day; 20 kg; ordered 165 mg every 8 h. Min 17 × 20 = 340 mg/day; max 33 × 20 = 660 mg/day. Ordered 165 × 24/8 = 495 mg/day: safe. Outside the range: hold, call the prescriber.

TRAP
mg/kg/day is a 24 h total: compare per day or per dose, like with like.

4 IV infusion rates

Pump: rate (mL/h) = V (mL) ÷ T (h). **Gravity:** drip (gtt/min) = V × DF ÷ (T × 60). DF (gtt/mL) is printed on the tubing: macrodrip 10, 15 or 20; microdrip 60. **60-drop shortcut:** with microdrip, gtt/min = mL/h. **End time** = start + V ÷ rate.

EXAMPLE Same order, pump vs gravity
1,000 mL over 8 h. Pump: 1,000 ÷ 8 = 125 mL/h.
Gravity, DF 20: 1,000 × 20 ÷ (8 × 60) ≈ 41.67 ⇒ **42 gtt/min**.

EXAMPLE End time and microdrip
750 mL at 100 mL/h from 09:00: 750 ÷ 100 = 7.5 h (7 h 30 min) → ends 16:30. Microdrip: 100 × 60 ÷ 60 = 100 gtt/min.

TRAP
gtt/min needs minutes: hours alone give a rate 60 times too high.

5 Body surface area (Mosteller)

BSA (m²) = $\sqrt{\text{height (cm)} \times \text{weight (kg)} \div 3{,}600}$

EXAMPLE BSA-based dose (Drug C)
150 cm, 54 kg: $\sqrt{150 \times 54 \div 3{,}600} = \sqrt{2.25}$ = 1.5 m². Ordered 92 mg/m²: 1.5 × 92 = 138 mg.

TRAP
Mosteller's 3,600 assumes cm and kg: convert in and lb first.

6 Reconstitution & strength

concentration = drug ÷ **final** volume on the label. % (w/v) = g per 100 mL: 0.9% = 0.9 × 1,000 ÷ 100 = 9 mg/mL. Ratio 1:1,000 = 1 g in 1,000 mL = 1 mg/mL.

EXAMPLE Powder vial (Drug D, fictional label)
1.4 g; label: add 6.2 mL to obtain 7 mL. 1.4 × 1,000 ÷ 7 = 200 mg/mL. Order 320 mg: 320 ÷ 200 = 1.6 mL (3 mL syringe, 0.1 mL marks).

TRAP
Powder adds volume: use the final 7 mL, not the 6.2 mL of diluent.

7 Titration & drips

C (mcg/mL) = mg in bag × 1,000 ÷ mL in bag. mL/h = mcg/kg/min × kg × 60 ÷ C. Reverse: mcg/kg/min = mL/h × C ÷ (kg × 60). Units: mL/h = (units/h) ÷ (units/mL).

EXAMPLE mcg/kg/min → mL/h (Drug E)
360 mg in 300 mL: 360 × 1,000 ÷ 300 = 1,200 mcg/mL. 4.5 mcg/kg/min, 76.4 kg (as recorded): 4.5 × 76.4 × 60 ÷ 1,200 = 17.19 ⇒ **17.2 mL/h**.

EXAMPLE units/kg/h → mL/h (Drug F)
30,000 units in 400 mL = 75 units/mL. 9.5 units/kg/h × 64 kg = 608 units/h; 608 ÷ 75 ≈ 8.107 ⇒ **8.1 mL/h**.

TRAP
Bag in mg, dose in mcg: × 1,000 first; per-minute doses need × 60.

8 Rounding & syringes

Round once, at the end

- Weight: record to 0.1 kg, then use as recorded.
- gtt/min: whole drops. mL/h: pump step (0.1 mL/h here; some: whole mL/h).
- Syringe: to its marks, 0.01 mL (1 mL size) or 0.1 mL (3 mL size).

TRAP
Rounding a concentration or intermediate early shifts the final rate.

9 Safe notation

Never write	Risk	Write
1.0 mg	Read as 10 mg (trailing zero)	1 mg
.5 mg	Read as 5 mg (no leading zero)	0.5 mg
U or u	Read as 0 or 4	unit(s)
IU	Read as IV or 10	international unit(s)
QD / QOD	Daily and every other day confused	daily / every other day
µg	Read as mg	mcg
cc	Read as u (units)	mL
MS, MSO4, MgSO4	Morphine and magnesium confused	full drug name
100000 units	Zeros miscounted	100,000 units

TRAP
A trailing zero or a missing leading zero makes a tenfold dose error.

10 Medication rights & orders

Rights (patient, drug, dose, route, time) need system safeguards. **Complete order:** patient name and ID, date/time, drug, dose with unit, route, frequency, weight if weight-based, signature.

CLINICAL NOTE
Unclear or incomplete order: clarify with the prescriber before giving. ISMP high-alert drips (IV heparin, insulin): independent double check per policy.

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